

1st SIT EXAMINATION QUESTION PAPER:	Year Long 2019
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Module Code:	CC6001NA
Module Title:	Advance Database System Development
Module Leader:	Rohit Panday (Islington College)

Date:	Week 31
Day / Evening:	Day
Start Time:	
Duration:	2 Hours

Exam Type:	Unseen , Closed
Materials supplied:	None
Materials permitted:	Writing equipment only
Warning:	Candidates are warned that possession of unauthorised materials in an examination is a serious assessment offence.

Instructions to candidates:	Please Note: Inclusive of this cover page, this examination paper consists of 5 pages. The student must complete four (4) out of five (5) questions. This examination accounts for 40% of your total module grades.
	You are to return this examination paper, BEFORE you leave the examination room.
	DO NOT TURN PAGE OVER UNTIL INSTRUCTED

Attempt Any Four Questions.

Question 1

The following relational schema specifies a part of a relational database for organization's information system:

Employee(E#, E_Name, Gender, DOB, Address, Tel, D#)
Department (D#, D_Name, L#)
Location (L#, Street, Zone, Country)
Job(E#, D#, Job_title, Salary)

The following assumptions are made:

- Employee job for any department is specified by the employee number (S#), department number (M#), job_title (e.g. 'manager, or 'Assistant'), and salary.

Answer the following questions:

(a) Identify the primary key, and foreign key(s) if applicable, for each of the relations above. Indicate clearly which relation contains which foreign key(s).

[5 Marks]

(b) Specify the correct sequence in which the above four relations should be created, and the order in which the relations should be dropped (removed).

[4 Marks]

(c) Express each of the following queries firstly in SQL and secondly in Relational Algebra:

(i) Find the names of all the department where people from Kathmandu is working.

[5 Marks]

(ii) Find the name of those employee who have assigned for the job 'Project Manager' in the 'Development' department.

[5 Marks]

(iii) Find the list of country names where 'Mr. David' is working as 'Manager'.

[6 Marks]

Question 2

(a) Discuss data replication, replicated database, and fully replicated database.

[2 Marks + 2 Marks + 2 Marks]

(b) Identify the three types of failure in database system.

[3 Marks]

(d) Transaction processing is a key component of a relational database system. Explain the concept of transaction, and provide a diagram to clearly depict the state transaction for transaction execution.

[3 Marks + 5 Marks]

Question 3

(a) Discuss the purpose of database concurrency and database recovery.

[4 Marks + 4 Marks]

(b) In the context of transaction processing discuss the concepts of

- Locking
- Commit
- Log File
- Transaction record

[8 Marks]

(c) During transaction processing, the system takes checkpoints at specified intervals. Assume that sometimes after a particular checkpoint was taken a system failure occurred. Consider the following transactions in the scenario:

T1 started and was completed before the checkpoint was taken.
T2 started before the checkpoint and was completed before the system failed.
T3 started before the checkpoint but did not complete when the system failed.
T4 started after the checkpoint and was completed before the system failed.
T5 started after the checkpoint but did not complete when the system failed.

(i) Draw a diagram to illustrate the above set of transactions.

[5 Marks]

(ii) For each of the above transactions, indicate which must be redone, which must be undone and which is unaffected by the failure.

[4 Marks]

Question 4

(a) Discuss the three levels of the ANSI-SPARC model, namely, external level, conceptual level and internal level. Illustrate your answer with a diagram.

[7 Marks + 5 Marks]

(b) Outline the main functions for the two levels of mapping associated with the ANSI/SPARC model.

[4 Marks + 4 Marks]

(c) Discuss the main difference between DDL and DML, and give one example SQL statement for each.

[5 Marks]

Question 5

(a) List and briefly outline the three concurrency problems which could occur in a multi-transaction database environment.

[1 Marks + 3 Marks + 3 Marks + 3 Marks]

(b) The following list represents the sequence of events in an interleaved execution of a set of transactions T1, T2, ... T24 in a concurrency system based on locking, where A, B, ...H are data items. Assume that FETCH A acquires an S lock on A, UPDATE A promotes that lock to an X lock, and all locks are held to the next synch point.

<i>Time</i>	<i>transaction</i>	<i>operation</i>
t0		
t1	T1	FETCH A
t2	T4	FETCH D
t3	T5	FETCH A
t4	T2	FETCH E
t5	T2	UPDATE E
t6	T3	FETCH F
t7	T2	FETCH F
t8	T1	COMMIT
t9	T6	FETCH A
t10	T5	ROLLBACK
t11	T6	FETCH C
t12	T6	UPDATE C
t13	T7	FETCH G
t14	T8	FETCH H
t15	T9	FETCH G
t16	T9	UPDATE G
t17	T8	FETCH E
t18	T7	COMMIT
t19	T9	FETCH H
t20	T3	FETCH G
t21	T10	FETCH E
t22	T9	UPDATE H
t23	T2	UPDATE F
t24		

(i) Provide a Wait-for-Graph to illustrate whether there is any deadlock at time *t24*. You are advised to provide full and appropriate workings.

[10 Marks]

(ii) Discuss how the system could recover if it were deadlocked at time *t25*, and justify your choice of victim transaction(s).

[5 Marks]